SIMATS ENGINEERING
ASSESSMENT TOOL 1 - ANALYTICAL PROBLEM SOLVING

Student Details	Assessment Details
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Course Outcome Covered: CO2 - Apply suitable data preprocessing techniques such as data cleaning, data integration, data transformation, and data reduction for effective data preparation (BL3, BL4).

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ANNEXURE B – SOLUTIONS

Q1. Equal-Frequency Binning (Equi-depth)

Given dataset (already sorted): 110, 115, 118, 120, 122, 125, 128, 130, 135, 138, 140, 145, 150, 155, 160.

1. Divide the data into three equal-frequency bins

Equal-frequency binning, also called equi-depth binning, divides sorted observations into groups that contain the same number of records. Here, the total number of observations is n = 15 and the required number of bins is k = 3. Therefore, the number of values in each bin is n/k = 15/3 = 5.

Bin	Original values	Mean calculation	Bin mean
Bin 1	110, 115, 118, 120, 122	(110 + 115 + 118 + 120 + 122) / 5	117.0
Bin 2	125, 128, 130, 135, 138	(125 + 128 + 130 + 135 + 138) / 5	131.2
Bin 3	140, 145, 150, 155, 160	(140 + 145 + 150 + 155 + 160) / 5	150.0

2. Smooth the values using bin means

In smoothing by bin means, each original value is replaced by the arithmetic mean of the bin to which it belongs. The calculation is performed separately for each bin, so every value within a bin receives the same replacement value.

Smoothed values by bin: Bin 1 becomes 117, 117, 117, 117, 117; Bin 2 becomes 131.2, 131.2, 131.2, 131.2, 131.2; and Bin 3 becomes 150, 150, 150, 150, 150. Therefore, the full smoothed dataset is 117, 117, 117, 117, 117, 131.2, 131.2, 131.2, 131.2, 131.2, 150, 150, 150, 150, 150.

3. Explain how binning reduces noise in medical data

Binning smooths random changes caused by sensor noise, movement, or measurement conditions by replacing values with their group mean. This makes broad patterns easier to see, but can hide clinically important extremes. Original readings must always be retained.

Q2. Normalization Techniques

Salaries (₹): 25,000; 30,000; 45,000; 60,000; 80,000; 1,20,000. Minimum = ₹25,000, maximum = ₹1,20,000, and range = ₹95,000. Normalization puts numerical values on a comparable scale.

1. Min-Max normalization to the range 0–1

Formula: $x' = (x - \text{minimum}) / (\text{maximum} - \text{minimum})$. For this dataset, $x' = (x - 25{,}000) / (1{,}20{,}000 - 25{,}000) = (x - 25{,}000) / 95{,}000$.

Salary (₹)	Min-Max value
25,000	(25,000 − 25,000) / 95,000 = 0.000
30,000	5,000 / 95,000 = 0.053
45,000	20,000 / 95,000 = 0.211
60,000	35,000 / 95,000 = 0.368
80,000	55,000 / 95,000 = 0.579
1,20,000	95,000 / 95,000 = 1.000

2. Z-score normalization

Z-scores show each salary's distance from the mean in standard-deviation units. The mean is $\mu = (25{,}000 + 30{,}000 + 45{,}000 + 60{,}000 + 80{,}000 + 1{,}20{,}000) / 6 = ₹60{,}000$.

Using the population standard deviation, $\sigma = \sqrt{[\Sigma(x - \mu)^2 / N]}$. The sum of squared deviations is 6,350,000,000, so $\sigma = \sqrt{(6{,}350{,}000{,}000 / 6)} = \sqrt{1{,}058{,}333{,}333.33} \approx ₹32{,}532.04$. The z-score formula is $z = (x - \mu) / \sigma$.

Salary (₹)	Z-score (approx.)
25,000	−1.076
30,000	−0.922
45,000	−0.461
60,000	0.000
80,000	0.615
1,20,000	1.844

Negative scores are below the mean, positive scores are above it, and zero corresponds to the mean.

3. Decimal scaling

Decimal scaling divides each value by 10^j, choosing the smallest j that makes the largest absolute scaled value less than 1. Here j = 6 because 120,000 / 10^6 = 0.12. Scaled values: 0.025, 0.030, 0.045, 0.060, 0.080, 0.120.

4. Compare the effects of the three methods

Comparison: Min-Max scales values to 0–1 but is sensitive to extremes; Z-score expresses distance from the mean in standard deviations but has no fixed range; decimal scaling reduces magnitude by shifting the decimal point without centring the data. The choice depends on the algorithm and dataset.

Q3. Covariance and Mean Analysis

The dataset gives the annual inflation rate x and stock-market return y over five years. Inflation values are 3.0, 4.5, 2.5, 5.0, and 3.5 (%); corresponding returns are 10, 15, 8, 18, and 12 (%). The mean provides the reference value for calculating how each observation deviates from its average.

1. Compute the means of x and y

Mean inflation rate: $\bar{x} = \Sigma x / n = (3.0 + 4.5 + 2.5 + 5.0 + 3.5) / 5 = 18.5 / 5 = 3.7\%$.

Mean market return: $\bar{y} = \Sigma y / n = (10 + 15 + 8 + 18 + 12) / 5 = 63 / 5 = 12.6\%$.

2. Calculate covariance

Covariance measures whether two variables tend to move together. For each year, subtract the corresponding mean from x and y, then multiply the two deviations. A positive product means both deviations have the same sign; a negative product means they have opposite signs.

Year	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
1	3.0 − 3.7 = −0.7	10 − 12.6 = −2.6	(−0.7)(−2.6) = 1.82
2	4.5 − 3.7 = 0.8	15 − 12.6 = 2.4	(0.8)(2.4) = 1.92
3	2.5 − 3.7 = −1.2	8 − 12.6 = −4.6	(−1.2)(−4.6) = 5.52
4	5.0 − 3.7 = 1.3	18 − 12.6 = 5.4	(1.3)(5.4) = 7.02
5	3.5 − 3.7 = −0.2	12 − 12.6 = −0.6	(−0.2)(−0.6) = 0.12
Total			16.40

Sample covariance: $s_{xy} = \Sigma[(x - \bar{x})(y - \bar{y})] / (n - 1) = 16.40 / (5 - 1) = 16.40 / 4 = 4.10$.

If the five observations represent the entire population rather than a sample, population covariance = $\Sigma[(x - \bar{x})(y - \bar{y})] / N = 16.40 / 5 = 3.28$.

3. Interpret the result

The covariance is positive under both conventions (sample covariance = 4.10; population covariance = 3.28). In this dataset, years with inflation above its mean also have market returns above their mean,

while years below the mean tend to have returns below the mean. Therefore, the variables show a positive direction of co-movement in these five observations.

Covariance alone does not provide a standardized measure of relationship strength because its magnitude depends on the units and scales of the variables. It also does not prove that inflation causes stock returns to change. More observations and additional economic factors would be needed for a reliable investment or economic conclusion.

Q4. Correlation Analysis

The stock prices are Company X = 200, 210, 220, 215, 225 and Company Y = 180, 190, 200, 195, 205 for times t1 to t5. Pearson's correlation coefficient, r, measures the direction and strength of a linear relationship between two numerical variables. Its value ranges from −1 to +1.

1. Compute the Pearson correlation coefficient

First calculate the means: $\bar{X} = (200 + 210 + 220 + 215 + 225) / 5 = 1{,}070 / 5 = 214$. $\bar{Y} = (180 + 190 + 200 + 195 + 205) / 5 = 970 / 5 = 194$.

Next, subtract the mean from each observation. Notice that Company Y is exactly 20 units lower than Company X at every time point, so their deviations from their respective means are identical.

Time	Company X	Company Y	dx = X − 214	dy = Y − 194
t1	200	180	−14	−14
t2	210	190	−4	−4
t3	220	200	6	6
t4	215	195	1	1
t5	225	205	11	11

Calculate the required sums: $\Sigma(dx \times dy) = 196 + 16 + 36 + 1 + 121 = 370$. Since dx and dy are identical for every observation, $\Sigma dx^2 = 370$ and $\Sigma dy^2 = 370$.

Pearson formula: $r = \Sigma(dx \times dy) / \sqrt{[\Sigma dx^2 \times \Sigma dy^2]}$. Substituting the values gives $r = 370 / \sqrt{(370 \times 370)} = 370 / 370 = +1.00$.

2. Interpret the strength and direction

The coefficient r = +1.00 indicates a perfect positive linear correlation in the supplied data. The prices rise and fall together in exactly the same pattern: whenever Company X changes by a particular amount, Company Y changes by the same amount. The relationship is positive because the two variables move in the same direction, and perfect because every pair lies exactly on a straight line.

3. Explain the effect on investment decisions

If the same pattern continued, holding both stocks would provide little diversification against the shared price movements, because their changes are perfectly correlated in this dataset. An investor

seeking diversification would normally examine assets with lower or negative correlations, along with risk, expected return, liquidity, and investment goals.

This result is based on only five observations and must not be treated as a guarantee of future performance. Correlation describes association, not causation, and relationships between stock prices can change over time. Actual investment decisions should use broader historical data and an assessment of the companies and market conditions.

Q5. Standardization and Deviation

Employee ages: 21, 23, 25, 28, 30, 32, 35, 40 years. MAD is the average absolute distance from the mean; a z-score measures distance from the mean in standard-deviation units.

1. Compute mean absolute deviation (MAD)

First calculate the mean age: $\bar{x} = \Sigma x / N = (21 + 23 + 25 + 28 + 30 + 32 + 35 + 40) / 8 = 234 / 8 = 29.25$ years.

Calculate each absolute deviation $|x - \bar{x}|$; absolute values prevent deviations above and below the mean from cancelling.

Age (x)	Absolute deviation $\|x - 29.25\|$
21	$\|21 - 29.25\| = 8.25$
23	$\|23 - 29.25\| = 6.25$
25	$\|25 - 29.25\| = 4.25$
28	$\|28 - 29.25\| = 1.25$
30	$\|30 - 29.25\| = 0.75$
32	$\|32 - 29.25\| = 2.75$
35	$\|35 - 29.25\| = 5.75$
40	$\|40 - 29.25\| = 10.75$
Total	40.00

$MAD = \Sigma|x - \bar{x}| / N = 40 / 8 = 5$ years. Thus, the average absolute distance from the mean age is 5 years.

2. Calculate z-scores for the first three ages

Treating the eight ages as the complete dataset, $\Sigma(x - \bar{x})^2 = 283.5$. Population variance = $283.5 / 8 = 35.4375$, so $\sigma = \sqrt{35.4375} \approx 5.953$ years.

Formula: $z = (x - \mu) / \sigma$, where μ is the mean and σ is the standard deviation.

Age	Deviation from mean	Z-score calculation	Z-score (approx.)
21	$21 - 29.25 = -8.25$	$-8.25 / 5.953$	-1.386
23	$23 - 29.25 = -6.25$	$-6.25 / 5.953$	-1.050
25	$25 - 29.25 = -4.25$	$-4.25 / 5.953$	-0.714

3. Interpret positive and negative z-scores

A positive z-score indicates an age above the mean, while a negative z-score indicates an age below the mean. A z-score of zero means the value equals the mean. Here, ages 21, 23, and 25 all have negative scores because they are below 29.25 years; for example, $z = -1.386$ means age 21 is about 1.386 standard deviations below the mean. MAD describes average absolute spread in years, whereas z-scores express each value's position in standard-deviation units.